TOPIC 3 · INTERACTIVE DASHBOARDS WITH CUSTOM SKILLS
LAB 8 THEME · INTERACTIVE FILTERS AND REFRESH

Lab 8 — Use the Dashboard Skill for Filters, Refresh and Safe Reruns

Maps to	LO3: build and validate an interactive Excel maintenance dashboard
Duration	35 minutes
Tools	Excel Copilot · Microsoft 365 Copilot web · OneDrive
Mock data	data/Northstar_Work_Orders.xlsx
Lab files	data/templates/water-service-dashboard/SKILL.md (Reviewed dashboard Skill for reruns)
Course	GenAI for Engineers · v2.3
Prompts	https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-8

Goal

Continue Lab 7 by making four dropdown selectors control every KPI and chart. Test a single site, a zero-denominator selection and a combined category/owner filter. Then append two synthetic work orders in a sandbox copy and prove the dashboard expands. The dashboard also has four table slicers; every tile and chart follows the dropdowns and the slicers together.

What You Will Do

Continue Lab 7 by making four dropdown selectors control every KPI and chart. Test a single site, a zero-denominator selection and a combined category/owner filter. Then append two synthetic work orders in a sandbox copy and prove the dashboard expands. The dashboard also has four table slicers; every tile and chart follows the dropdowns and the slicers together.

What You Will Build

Northstar_Dashboard.xlsx with four working filters plus Northstar_Dashboard_Refresh_Test.xlsx, filter evidence and a verified baseline snapshot for PowerPoint.

Prerequisites

- Microsoft 365 account with Copilot available in Word, Excel, PowerPoint and Outlook. Availability depends on the assigned licence and tenant settings; confirm access with the trainer before class.
- Desktop Excel on Windows or Mac is recommended; save working copies to your assigned OneDrive course folder with AutoSave on. Do not overwrite the supplied input.
- Download this whole lab folder. All files are synthetic; work-genai is your own learner output folder.
- Lab 7 dashboard or supplied formula checkpoint.
- Lab 7 reviewed water-service-dashboard Skill installed or explicitly labelled manual route.

Data note

Northstar Water Services is a fictional water-pipe maintenance and service company. All service hubs, staff and work orders are synthetic. Use the fixed reporting date 2026-09-30; missing facts remain UNKNOWN. Proposals do not change real assignments. Training email drafts are never sent.

Steps

Step 1. Open your Lab 7 baseline dashboard. Rejoin using data/checkpoints/Northstar_Verified_Checkpoint.xlsx only if needed; record that it is an example formula dashboard. Set all four selections B2:B5 to All and confirm 120. Save baseline before creating any refresh-test copy.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@water-service-dashboard Before testing filters, report the baseline Northstar Water Services dashboard source, date and scope. Confirm 120 unique work orders and all four selectors All. Identify the baseline file that must remain untouched when making the refresh-test copy.
```

Step 2. Use Data > Data Validation > Allow List to set dropdowns on B2 Site, B3 Job_Category, B4 Status and B5 Assigned_To. Include All plus the source values, using Filter_Lists if supplied. Ask Copilot to apply all four selections to EVERY KPI and chart summary. A selection of All bypasses only that filter; selected dimensions combine with AND.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@water-service-dashboard Make Dashboard B2:B5 control all six KPIs and three chart summaries with AND logic. All bypasses its dimension. Use COUNTIFS/SUMIFS or SUMPRODUCT with structured table references. Blank owner means unassigned and must not disappear from All. Keep N/A for no closed jobs.

EXCEL - ALLOW EDITING:
@water-service-dashboard Apply the reviewed filter plan and dropdown lists. Keep existing source rows and chart positions. Show the selected scope beside the reporting date and make all charts recalculate from the same selected rows.
```

Step 3. Choose Woodlands Hub in B2 and leave other selectors All. Then set B2 back to All and click Woodlands Hub in the Site slicer instead: the D3 banner reports 24 of 120 rows and every tile must show the same values. Clear the slicer afterwards with its Clear Filter button. Record evidence in the Evidence_Log sheet.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@water-service-dashboard For the dashboard selection Site = Woodlands Hub and all other dimensions All, list the expected six KPI values from tblWorkOrders and verify each chart sum matches the selected total. Confirm the Site slicer gives the same six values and that D3 reports the slicer filter.
```

Step 4. Set Status to New with other selectors All. Confirm zero closed -> N/A. Then set Site = Woodlands Hub, Job_Category = Leak Repair, Assigned_To = Daniel Lee, Status = All, a combination with no rows; total, backlog, overdue, closed and cost must be 0 and rate N/A. Reset all four to All and verify baseline controls. Repeat the Status = New case with the Status slicer instead of B4 and confirm the same N/A; then clear every slicer.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@water-service-dashboard Check two dashboard edge cases without changing source data: Status New with other selectors All; then Site Woodlands Hub, Job_Category Leak Repair, Assigned_To Daniel Lee, Status All. Verify zero closed gives N/A and no selected rows gives zero count/cost with N/A rate. Show how all charts follow the same selection. Restore all selectors to All.
```

Step 5. Save a COPY as Northstar_Dashboard_Refresh_Test.xlsx. In the last row of tblWorkOrders press Tab to append, then copy the two rows from the Refresh_Test_Rows sheet (they already hold real Excel dates). WO-2026-0121 is Assigned, due 2026-09-29 and cost 150; WO-2026-0122 is Closed on time and cost 250. No existing IDs are edited.

Step 6. Recalculate with all selections All. If using PivotTables also Data > Refresh All and inspect source extent. Verify total122, closed49, backlog61, overdue baseline+1, cost baseline+400 and on-time numerator baseline+1 over49. If formulas do not expand, fix structured references/table extent. Keep this sandbox copy as refresh evidence; do not use its figures in the baseline presentation.

PROMPT / CHECK — copy and adapt

```
EXCEL - CHAT ONLY:
@water-service-dashboard Verify the sandbox append test for the two new work orders. Reconcile total 122, closed 49, backlog 61, overdue increased by 1, cost increased by S$400 and on-time numerator increased by 1. Confirm no original source record changed and all three charts total 122 with All selections. Compare original IDs, statuses, owners, dates and costs to the saved baseline; all original 120 rows must match.
```

Step 7. In the sandbox workbook request a controlled Skill rerun after the append. Review the named replacement areas, then approve only those areas. Verify one dashboard, three charts, original 120 source rows unchanged and both legitimate new rows retained. Keep the 120-row baseline separate for Lab 9.

PROMPT / CHECK — copy and adapt

```
EXCEL - PLAN:
@water-service-dashboard Refresh the existing dashboard for the sandbox 122-row table and same reporting date. Preserve all original source records, both new rows, .Rules, Pareto_* and unrelated sheets. Identify named output replacements and ask before editing; keep one dashboard and three charts.

EXCEL - ALLOW EDITING:
@water-service-dashboard Apply only the reviewed sandbox dashboard rerun. Retain the chosen source/date and make every KPI and chart follow the four filters; reconcile 122 total,61 backlog,51 overdue,49 Closed,31/49 on-time and S$36,227.50. Do not alter the untouched baseline workbook.
```

Step 8. Return to the untouched 120-row Northstar_Dashboard.xlsx. Set selectors All and record a frozen evidence snapshot dated 2026-09-30 for Lab 9. Copy data/templates/Northstar_Operations_Brief_Template.docx to Northstar_Verified_Snapshot.docx in Word and insert the reviewed summary and chart values. Save chart images if needed with right-click chart > Save as Picture; the slide copy will be a static snapshot, not a live dashboard.

PROMPT / CHECK — copy and adapt

```
WORD COPILOT:
Using only the verified baseline dashboard evidence I paste here, prepare the frozen Northstar Water Services presentation snapshot. Include reporting date, All scope, 120-row baseline, six KPIs, chart values, source workbook link and limitations. Exclude the 122-row sandbox. Keep proposed priorities separate from observed facts.
```

Sample Prompts

Use the app labelled for each prompt (or adapt it). Name the table and columns, state the output, and always verify the result. Use the linked prompts web page for one-click Copy buttons.

Copy-ready prompts: https://tertiarycourses.github.io/IES-GenAI-for-Engineers/prompts/#lab-8

#	Purpose	Prompt
1	Scope	State actual source, report date and selected scope before acting.
2	Verify	Explain the source/formula checks and preserve unknowns and original records.

Test It

Check	Expected result
Woodlands control	Total 24; backlog 8; overdue 7; closed 16; cost S$6,668.50; on-time 10/16.
Interactive scope	Changing any dropdown or slicer affects all six KPI tiles and all chart counts; D3 reports slicer filters.
Edge case	No Closed -> N/A; no rows -> zero count/cost and N/A rate.
Refresh sandbox	122 total;49 closed;61 backlog;51 overdue;S$36,227.50;31/49 on-time.
Handoff baseline	Return to 120-row untouched baseline; date and All scope recorded.

Checkpoint

If you fell behind, use data/checkpoints/Northstar_Verified_Checkpoint.xlsx and data/checkpoints/Northstar_Verified_Snapshot.docx. Record checkpoint used in Prompt_Log; then perform this lab, rather than claiming the checkpoint as your output.

Troubleshooting

If this happens	Fix
Copilot cannot find the source	Use the app file picker to select the actual OneDrive file. A typed local path or a different app chat is not shared context. If needed copy only the verified snapshot into the current app.
A total disagrees with Control	Clear dashboard filters to All, check table extent and duplicate IDs, recalculate, then compare definitions and reporting date. Reconcile before continuing.
The named feature is missing	Confirm the correct assigned account, English display language and app version. Use the stated manual practice route; record feature unavailable. This is not evidence that a native skill ran.

Challenge

Repeat the task for one different site and explain how the filter scope changes the result. Keep the original verified output.

Reflection

Which evidence proves the output is correct, and which decision still needs a person?